Algebraic Coding Theory

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Abstract

Notes taken while studying Algebraic Coding Theory, mostly from Hill book [1] and Lint book [2].

Usually while reading books and papers I take handwritten notes in a notebook, this document contains some of them re-written to $LaTeX$.

The proofs may slightly differ from the ones from the books, since I try to extend them for a deeper understanding.

Contents

1 Vector spaces over finite fields, quick recap

Let q be a prime power.

$V(n,q)$: set $GF(q)^n$ of all ordered n-tuples over $GF(q)$. $V(n,q)$ is a vector space.

A set of vectors $\{v_1, v_2, \ldots, v_r\}$ is said to be *linear dependent* if $\exists$ scalars $a_1, \ldots, a_r$, not all zero, such that

$$a_1v_1 + a_2v_2 + \ldots + a_rv_r = 0$$

They are *linear independent* if

$$a_1v_1 + a_2v_2 + \ldots + a_rv_r = 0 \overset{implies}{\Longrightarrow} a_1 = a_2 = \ldots = a_r = 0$$

Let C a subspace of $V(n,q)$. Then a subset $\{v_1, v_2, \ldots, v_r\} \subseteq C$ is called a *generating set* (or *spanning set*) of C if every vector in C can be expressed as a linear combination of $v_1, \ldots, v_r$.

A generating set of C which is also linearly independent is called a *basis* of C.

Theorem 4.3. Suppose $\{v_1, v_2, \ldots, v_k\}$ is a basis of a subspace C of $V(n,q)$. Then

i. every vector of C can be expressed uniquely as a linear combination of the basis vectors.

ii. C contains exactly q^k vectors.

Proof. i. suppose vector $x \in C$ such that

$$x = a_1v_1 + a_2v_2 + \cdots + a_kv_k$$
$$x = b_1v_1 + b_2v_2 + \cdots + b_kv_k.$$

then

$$(a_1 - b_1)v_1 + (a_2 - b_2)v_2 + \cdots + (a_k - b_k)v_k = 0.$$

But the set $\{v_1, v_2, \ldots, v_k\}$ is linearly independent, and so

$$a_i - b_i = 0 \quad \text{for } i = 1, 2, \ldots, k,$$

i.e.,

$$a_i = b_i \quad \text{for } i = 1, 2, \ldots, k.$$

ii. By (i), the q^k vectors

$$\sum_{i=1}^{k} a_iv_i \qquad (a_i \in \mathrm{GF}(q))$$

are precisely the distinct vectors of C.

□

2 Linear codes

The theorems of this section are from [1] and follow the original enumeration of the book.

Note about notation:
A q-ary linear code $[n, k, d]$-code is also a q-ary (n, q^k, d)-code, but the opposite is not always true.

2.1 Some definitions

Let q be a prime power. View $(F_q)^n$ as the vector space $V(n, q)$.

Definition (Linear code over GF(q)). A *linear code over* $GF(q)$ is a subspace of $V(n, q)$, with $n \geq 0$.

Thus $C \subseteq V(n, q)$ is a linear code iff

i. $u + v \in C \quad \forall u, v \in C$

ii. $au \in C \quad \forall u \in C,\ a \in GF(q)$

If C is a k-dimensional subspace of $V(n, q)$, then the linear code C is called an $[n, k]$-code. (sometimes with minimum distance d: $[n, k, d]$-code).

Definition (Weight). Weight, $w(x)$, of a vector $x \in V(n, q)$ is the number of non-zero entries of x.

Lemma 5.1. if $x, y \in V(n, q)$, then $d(x, y) = w(x - y)$.

Proof. the vector $x - y$ has non-zero entries in preceisely those places where x and y differ. □

Theorem 5.2. let C a linear code, let $w(C)$ be the smallest of weights of the non-zero codewords of C. Then $d(C) = w(C)$.

Proof. $\exists\ x, y \in C$ s.th. $d(C) = d(x, y)$.
By Lemma 5.1, $d(C) = w(x - y) \geq w(C)$, since $x - y$ is a codeword of the linear code C (ie. $x - y \in C$).
On the other hand, for $x \in C$,
$w(C) = w(x) = d(x, 0) \geq d(C)$, since $0 \in C$.
Hence, $d(C) \geq w(C)$ and $w(C) \geq d(C) \implies d(C) = w(C)$. □

Definition (Generator matrix (GM)). a $k \times n$ matrix whose rows form a basis of a linear $[n, k]$-code is called a *generator matrix* (GM) of the code.

2.2 Equivalence

Theorem 5.4. two known $k \times n$ matrices generate equivalent linear $[n, k]$-codes over $GF(q)$ if one matrix can be obtained from the other by a sequence of operations of the following types:

R1. permutation of the rows

R2. multiplication of a row by a non-zero scalar

R3. addition of a scalar multiple of one row to another

C1. permutation of the columns

C2. multiplication of any column by a non-zero scalar

Theorem 5.5. let G a GM of an $[n, k]$-code. Then by performing the previous operations, G can be transformed to the standard form $[I_k|A]$ where I_k is the $k \times k$ identity matrix, and A is a $k \times (n-k)$ matrix.

2.3 Encoding with a linear code

Let C a $[n, k]$-code over $GF(q)$ with GM G. C contains q^k codewords $\Longrightarrow$ can be used to communicate any one of q^k distinct messages.

Iddentify these messages with the q^k tuples of $V(k, q)$.

Encoding:

msg vector: $u = u_1 u_2 \dots u_k$; rows of G: $r_1 r_2 \dots r_k$. Encode

$$uG = \sum_{i=1}^{k} u_i r_i \in C$$

ie. uG is a codeword of C; a linear combination of the rows of the GM.

Encoding function:

$$\begin{aligned} V(k,q) &\longrightarrow C \subseteq V(n,q) \\ u &\longmapsto uG \end{aligned}$$

If G is in standard form (5.5), $G = [I_k|A]$ with $A = [a_{ij}]$, then encoding u is:

$$x = uG = x_1 x_2 \dots x_k x_{k+1} \dots x_n$$

where

$$\begin{aligned} x_i &= u_i \quad \text{for } 1 \leq i \leq k, \text{ message digits} \\ x_{k+i} &= \sum_{j=1}^{k} a_j i u_j \quad \text{for } 1 \leq i \leq n-k, \text{ check digits} \end{aligned}$$

the check digits add redundancy to gie protection against noise.

2.4 Decoding with a linear code

Codeword $x = x_1x_2\dots x_n$, received vector $y = y_1y_2\dots y_n$, error vector $e = y - x = e_1e_1\dots e_n$.

2.4.1 Cosets

Cosets here are similar to the ones from typical group theory, just from another perspective.

Definition (Coset)**.** Let C a $[n,k]$-code over $GF(q)$ and $a \in V(n,q)$. Then the set $a + C = \{a + x | x \in C\}$ is called a *coset* of C.

Definition (Coset leader)**.** the vector having minimum weight in a coset is called the *coset leader*.

Lemma 6.3. Let $a + C$ a coset of C and $b \in a + C$. Then $b + C = a + C$.

Proof. since $b \in a + C \implies b = a + x$ for some $x \in C$. If $b + y \in b + C$ then

$$b + y = (a + x) + y = a + (x + y) \in a + C$$

For the other direction, if $a + x \in a + C$ then

$$a + x = (b - x) + z = b + (z - x) \in b + C$$

hence $a + C \subseteq b + C$.
Therefore $b + C = a + C$. □

Theorem 6.4 (Lagrange)**.** suppose $C \subseteq V(n,q)$ is a $[n,k]$-code over $GF(q)$. Then

i. evvery vector of $V(n,q)$ is in some coset of C

ii. every coset contains exactly q^k vectors

iii. two cosets either are disjoint or coincide (partial overlap is impossible)

Proof. i. if $a \in V(n,q)$ then $a = a + 0 \in a + C$

ii. the map

$$\begin{aligned} C &\longrightarrow a + C \ \forall\ x \in C \\ x &\longmapsto a + x \end{aligned}$$

is a one-to-one map, hence $|a + C| = |C| = q^k$.

iii. by contradiction, suppose cosets $a+C$, $b+C$ overlap. Then for some vector v,

$$v \in (a + C) \cap (b + C)$$

thus for some $x, y \in C$, $v = a + x = b + y$.
Hence $b = a + (x - y) \in a + C$, thus by Lemma 6.3 we have $b + C = a + C$. □

Corollary . 6.4 shows that $V(n,q)$ is partitioned into disjoint cosets of C:

$$V(n,q) = (0+C) \cup (a_1+C) \cup \ldots \cup (a_s+C)$$

where $s = q^{n-k}-1$, and by Lemma 6.3 we may take $0, a_1, \ldots, a_s$ to be coset leaders.

Example 6.5. Let C be the binary $[4,2]$-code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

ie. $C = \{0000,\ 1011,\ 0101,\ 1110\}$.
Then the cosets of C are

$$\begin{aligned}
&0000 + C = C \text{ itself} \\
&1000 + C = \{1000,\ 0011,\ 1101,\ 0110\} \\
&0100 + C = \{0100,\ 1111,\ 0001,\ 1010\} = 0001 + C \\
&0010 + C = \{0010,\ 1001,\ 0111,\ 1100\}
\end{aligned}$$

by Lemma 6.3: $0001 \in 0100 + C$, thus $0001 + C = 0100 + C$.

2.5 Probability of error correction (in binary linear codes)

To evaluate how good a code is, must calculate the *probability* that a received vector will be decoded as the codeword which was sent.

Let p be the symbol error probability.

Probability that the error vector is agiven vector of weight i is $p^i(1-p)n-i$.

Theorem 6.7. Let C a binary $[n,k]$-code, and for $i = 0, 1, \ldots, n$ let α_i denote the number of coset leaders of weight i.

Then the probability $P_{corr}(C)$ that a received vector decoded by means of a standard array is the codeword which was sent is

$$P_{corr}(C) = \sum_{i=0}^{n} \alpha_i \cdot p^i (1-p)^{n-i}$$

Example 6.8. $[n,k]$-code from 6.5, coset leaders are $0000, 1000, 0100, 0010$. Hence $\alpha_0 = 1,\ \alpha_1 = 3,\ \alpha_2 = \alpha_3 = \alpha_4 = 0$, so

$$P_{corr}(C) = (1-p)^4 + 3p(1-p)^3 = (1-p)^3(1+2p)$$

Remark 6.9. if $d(C) = 2t+1$ or $d(C) = 2t+2$, then C can correct any t errors.

Hence every vector of weight $\leq t$ is a coset leader and so $\alpha_i = \binom{n}{i}$ for $0 \leq i \leq t$.

Definition (Rate)**.** $[n,k]$-code C uses n symbols to send k message symbols; it has *rate* $R(C) = k/n$.

A good code will have a high rate.

Definition (Capacity)**.** the *capacity* $\mathscr{C}(p)$ of a binary symmetric channel with symbol error probability p is

$$\mathscr{C}(p) = 1 + p \cdot \log_2(p) + (1-p) \cdot \log_2(1-p)$$

Theorem 6.12 (Shanon's theorem)**.** channel binary symmetric with symbol error probability p.

Let $R < \mathscr{C}(p)$.

Then for any $\epsilon > 0$, $\exists$ for sufficient large n, an $[n,k]$-code C of rate $k/n \geq R$ such that $P_{err}(C) < \epsilon$.

Definition (P_{symb})**.** P_{symb} denotes the average probability that a message symbol is in error after decoding.

Theorem 6.14. let C a binary [n,k]-code and let A_i denote the number of codewords of C of weight i.

Then, if C is used for error detection, the probability of an incorrect message being undetected is

$$P_{undetec}(C) = \sum_{i=1}^{n} A_i \cdot p^i (1-p)^{n-i}$$

Example 6.15. for the code of 6.5,

$$P_{undetec}(C) = \underbrace{1 \cdot p^2(1-p)^2}_{0101} + \underbrace{2 \cdot p^3(1-p)^{4-3=1}}_{1011,\ 1110} = p^2 - p^4$$

2.6 Exercises

Exercise 5.1. is the binary (11, 24, 5)-code linear?

Proof. recall: a q-ary linear $[n,k,d]$-code is also a q-ary (n, q^k, d)-code, but the opposite is not always true.

$\Longrightarrow$ $(11, 24, 5)$-code to be linear it would mean that $24 = q^k$.

Since it is binary, $q = 2$. But $24 \neq 2^k$, so it can not be linear. □

Exercise 5.2. E_n is the code of all even-weight vectors of $V(n,2)$, which is linear. What are the parameters $[n,k,d]$ of E_n? Write a GM for E_n in standard form.

Proof.

$$E_n = \{x \in V(n,2) | w(x) \text{ is even}\}$$

$\longrightarrow$ is the binary even-weight ode (the single parity-check code);
it's linear since

- the zero vector has even weight

- over $GF(2)$, the sum of two even-weight vectors has even weight

Recall: $F_q^n \iff V(n,q)$, so $V(n,2) \cong F_2^n$

$$\implies E_n = \{(x_1, \ldots, x_n) \in F_2^n | x_1 + \ldots + x_n = 0\}$$

minimum distance is 2, since

- no word of weight 1 is in E_n
- words of weight 2 are in E_n

Therefore, parameters are $[n, n-1, 2]$
GM:

$$G = [I_{n-1}|1]$$

which generates vectors of the form

$$(a_1, \ldots, a_{n-1}, a_1 + \ldots + a_{n-1})$$

□

Exercise 5.5. Prove that in a *binary linear code*, either all the codewords have even weight or exactly half have even weight and half odd weight.

Proof. let $C_e,\ C_o$ be subsets of C with even and odd weights respectively.
So that $C = C_e \cup C_o$ (disjoint union).
For binary vectors x, y:

$$w(x+y) = w(x) + w(y) \pmod 2$$

ie. over F_2, the parity of the weight is additive:

$$\begin{aligned} even + even &= even \\ even + odd &= odd \\ odd + odd &= even \end{aligned}$$

Now we have two possible cases:

- C has no odd weighted codewords:
  $C_0 = \emptyset,\ \forall x \in C\ x$ is even weighted
- C has at least one odd-weighted codeword:
  let $u \in C_o$, consider the map

$$\begin{aligned} \phi : C_e &\longrightarrow C_o \\ c &\longmapsto c + u \end{aligned}$$

  with $c \in C_e,\ u \in C_o,\ \phi(c) \in C_o$.

  ϕ is bijective, since $\phi(\phi(c)) = (c+u) + u = c$,

  so ϕ is it's own inverse; hence $|C_e| = |C_o|$.

  Since $C = C_e \cup C_e$, half codewords are even weighted and half odd.

□

Exercise 6.1. construct standard arrays for codes having each of the GMs:

$$G_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad G_2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

Proof. recall: for a binary code with GM G, the code C consists of all binary linear combinations of the rows of G.

For G_1:
rows: $r_1 = (1,0),\ r_2 = (0,1))$. A codeword is any linear combination:

$$uG_1 = a \cdot r_1 + b \cdot r_2, \quad \text{with } a, b \in \{0,1\}$$

4 combinations:

$$\begin{aligned} 0 \cdot r_1 + 0 \cdot r_2 &= (0,0) \\ 1 \cdot r_1 + 0 \cdot r_2 &= (1,0) \\ 0 \cdot r_1 + 1 \cdot r_2 &= (0,1) \\ 1 \cdot r_1 + 1 \cdot r_2 &= (1,0) + (0,1) = (1,1) \end{aligned}$$

$$\implies C_1 = \{00, 10, 01, 11\}$$

standard array: 00 01 10 11

For G_2:
rows: $r_1 = (1,0,1),\ r_2 = (0,1,1)$. Codeword: $a \cdot r_1 + b \cdot r_2$ with $a, b \in \{0,1\}$.
4 combinations:

$$\begin{aligned} 0 \cdot r_1 + 0 \cdot r_2 &= (0,0,0) \\ 1 \cdot r_1 + 0 \cdot r_2 &= (1,0,1) \\ 0 \cdot r_1 + 1 \cdot r_2 &= (0,1,1) \\ 1 \cdot r_1 + 1 \cdot r_2 &= (1,0,1) + (0,1,1) = (1,1,0) \end{aligned}$$

$$\implies C_2 = \{000, 101, 011, 110\}$$

$\implies$ so the 1st row of the standard array is: 000 101 011 110.
Now, pick a non used vector: 001, add it to the code C_2:

$$\begin{aligned} &001 + C_{2_i} \\ &001 + 000 = 001 \\ &001 + 101 = 100 \\ &001 + 011 = 010 \\ &001 + 110 = 111 \end{aligned}$$

$\Longrightarrow$ 2nd row of the standard array: 001 100 010 111

Thus the standard array is

$$\begin{array}{c} 000\ 101\ 011\ 110 \\ 001\ 100\ 010\ 111 \end{array}$$

ie. $2^3 = 8$ vectors, a $V(3,2)$ vector space. $\square$

Exercise 6.4. C a binary $[n,k]$-code with minimum distance $2t+1$ (or $2t+2$). Given that p is very small, show that an approximate value of $P_{err}(C)$ is $\left(\binom{n}{t+1} - \alpha_{t+1}\right) p^{t+1}$, where α_{t+1} is the number of coset leaders of C of weight $t+1$.

Proof. From 6.9, if $d(C) = 2t+1$ (or $2t+2$) *Longrightarrow* C can correct any t errors.

Hence every vector of weight $\leq t$ is a coset leader and so $\alpha_i = \binom{n}{i}$ for $0 \leq i \leq t$.

$$P_{corr}(C) = \sum_{i=0}^{n} \alpha_i \cdot p^i (1-p)^{n-i} \tag{1}$$

where α_i is the number of coset leaders of weight i

From the $\binom{n}{i}$ vectors of weight i, exactly α_i are coset leaders, so the non coset leaders are $\binom{n}{i} - \alpha_i$. This is when a decoding error happens.

$$\Longrightarrow \; P_{err}(C) = \sum_{i=0}^{n} \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i}$$

Alternatively from 1:

$$P_{err}(C) = 1 - P_{corr}(C) = 1 - \sum_{i=0}^{n} \alpha_i \cdot p^i (1-p)^{n-i}$$

Notice that

$$\sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} = (p + (1-p))^n = 1^n = 1$$

thus

$$\begin{aligned} P_{err}(C) &= \sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} - \sum_{i=0}^{n} \alpha_i \cdot p^i (1-p)^{n-i} \\ &= \sum_{i=0}^{n} \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i} \end{aligned}$$

Now, from 6.9, $\forall i \le t,\ \alpha_i = \binom{n}{i}$, thus $\binom{n}{i} - \alpha_i = 0$ for the first t terms in the sum.

Thus

$$\begin{aligned} P_{err}(C) &= \sum_{i=0}^{n} \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i} \\ &= \left(\binom{n}{t+1} - \alpha_{t+1} \right) p^{t+1} (1-p)^{n-t+1} + \ldots \end{aligned}$$

where the $\ldots$ represent the terms with p^{t+2} and higher.

Then, since p is very small (by the exercise definition), $(1-p)^{n-t+1} \equiv 1$, and the other higher powers of p are negligible.

Hence,

$$P_{err}(C) = \left(\binom{n}{t+1} - \alpha_{t+1} \right) \cdot p^{t+1}$$

□

3 Dual code, PCM and Syndrome

3.1 Dual code

Definition (Orthogonal)**.** Inner product:

$$u \cdot v = u_1 v_1 + \ldots + u_n \cdot v_n = \sum_{i=1}^{n} u_i v_i$$

If $u \cdot v = 0$, then u and v are called *orthogonal*.

Lemma 7.1. for any $u, v, w \in V(n, q)$ and $\lambda, \mu \in GF(q)$,

i. $u \cdot v = v \cdot u$

ii. $(\lambda u + \mu v) \cdot w = \lambda(u \cdot w) + \mu(v \cdot w)$

Proof. (at exercise 7.1). □

Definition (Dual code)**.** given a linear $[n, k]$-code C, the *dual code* of C, denoted $C^\perp$, is defined to be the set of those vectors of $V(n, q)$ which are orthogonal to every codeword of C. ie.

$$C^\perp = \{ v \in V(n, q) \mid v \cdot u = 0 \ \forall\ u \in C \}$$

Lemma 7.2. C a $[n, k]$-code with GM G. Then $v \in V(n, q)$ belongs to $C^\perp$ iff v is orthogonal to every row of G, ie.

$$v \in C^\perp \Longleftrightarrow v \cdot G^T = 0$$

Proof. let rows of G be $r_1, \dots, r_k$, suppose $v \cdot r_i = 0 \ \forall i$ (ie. v orthogonal to every row of G).

If $u \in C$ (ie. a codeword of C), then $u = \sum_{i=1}^{k} \lambda_i r_i$ for some scalars λ_i, so that

$$v \cdot u = v \cdot \sum_{i=1}^{k} \lambda_i r_i = \sum_{i=1}^{k} \lambda_i \cdot \underbrace{(v \cdot r_i)}_{0} = \sum \lambda_i \cdot 0 = 0$$

$\implies$ hence v is orthogonal to every codeword of C, and so is in $C^{\perp}$. $\square$

Theorem 7.3. C a $[n, k]$-code over $GF(q)$. Then the dual code $C^{\perp}$ of C is a linear $[n, n-k]$-code.

Proof. i. first show that $C^{\perp}$ is a linear code:

let $v_1, v_2 \in C^{\perp}$ and $\lambda, \mu \in GF(q)$. Then $\forall u \in C$,

$$\begin{aligned}(\lambda v_1 + \mu v_2) \cdot u &= \lambda(v_1 \cdot u) + \mu(v_2 \cdot u) \\ &\text{notice that } v_1, v_2 \in C^{\perp}\text{, so} \\ &\text{that } v_1, v_2 \text{ are orthogonal with } u \\ &\text{hence } (v_1 \cdot u) \text{ and } (v_2 \cdot u) \text{ are } 0 \\ &= \lambda \cdot 0 + \mu \cdot 0 = 0\end{aligned}$$

Hence $\lambda \cdot v_1 + \mu \cdot v_2 \in C^{\perp}$, and so $C^{\perp}$ is linear, since it is closed under linear combinations.

ii. Next, show that $C^{\perp}$ has dimension $n - k$:

Let $G = [g_{ij}]$ be a GM of C.

Then by Lemma 7.2, the elements of $C^{\perp}$ are the vectors $v = (v_1, v_2, \dots, v_n)$, since $v \in C^{\perp} \iff v \cdot G^T = 0$, satisfying

$$\sum_{j=1}^{m} g_{ij} v_j = 0, \ \forall \ i = [k]$$

$\longrightarrow$ this is a system of k independent homogeneous equations in n unknowns, with solution space $C^{\perp}$. From linear algebra we know that it has dimension $n - k$. To see it:

if codes C_1 and C_2 are equivalent, then $C_1^{\perp}$, $C_2^{\perp}$ are also equivalent. Hence is enough to show that $\dim(C^{\perp}) = n - k$, where C is a standard form GM:

$$C = \begin{pmatrix} 1 & \dots & 0 & a_{1,1} & \dots & a_{1,n-k} \\ \vdots & & \vdots & \vdots & & \vdots \\ 0 & \dots & 1 & a_{k,1} & \dots & a_{k,n-k} \end{pmatrix}$$

Then

$$C^{\perp} = \{(v_1, \dots, v_n) \in V(n,q) \bigm| v_i + \sum_{j=1}^{n-k} a_{i,j} v_{k+j} = 0,\ i = 1, 2, \dots, k\}$$

For each of the q^{n-k} choices of $(v_{k+1}, \dots, v_n)$, $\exists!\ (v_1, v_2, \dots, v_n) \in C^{\perp}$.
Hence $|C^{\perp}| = q^{n-k}$, therefore $dim(C^{\perp}) = n - k$.

$\square$

Theorem 7.5. for any $[n,k]$-code C, $(C^{\perp})^{\perp} = C$.

Proof. clearly $C \subseteq (C^{\perp})^{\perp}$, since every vector in C is orthogonal to every vector in $C^{\perp}$.
But $dim((C^{\perp})^{\perp}) = n - (n-k) = k = dim(C)$, therefore $C = (C^{\perp})^{\perp}$. $\square$

3.2 PCM (parity-check matrix)

Definition (PCM)**.** A *parity-check matrix* (PCM) H for a $[n,k]$-code C is a generator matrix of $C^{\perp}$.
$\implies$ H is a $(n-k) \times n$ matrix satisfying $GH^T = 0$.

From 7.2 and 7.5,
if H a PCM of C, then

$$C = \{x \in V(n,q) \mid xH^T = 0\}$$

$\implies$ Any linear code is completely specified by a PCM.

Intuition . i. PCM: G specifies how to build a codeword, the PCM H specifies the rules a vector must follow to be considered a valid codeword.

ii. dual-code: orthogonal "shadow", see C as a plane through 3D origin; $C^{\perp}$ is the (normal vector) line emerging from the plane.

Rows of the PCM are the basis for the dual-code $C^{\perp} \implies$ ie. the rules that define C are themselves a code.

Theorem 7.6. if $G = [I_k|A]$ is the standard form generator matrix of an $[n,k]$-code C, then a PCM for C is $H = [-A^T \mid I_{n-k}]$.
eg.

$$G = \begin{bmatrix} 1 & \dots & 0 & a_{1,1} & \dots & a_{1,n-k} \\ \vdots & & \vdots & \vdots & & \vdots \\ 0 & \dots & 1 & a_{k,1} & \dots & a_{k,n-k} \end{bmatrix}$$

$$H = \begin{bmatrix} -a_{1,1} & \dots & -a_{1,n-k} & 1 & \dots & 0 \\ \vdots & & \vdots & \vdots & & \vdots \\ -a_{k,1} & \dots & -a_{k,n-k} & 0 & \dots & 1 \end{bmatrix}$$

which, in binary, the minus signs are unnecessary.

Definition (Standard form)**.** a PCM that is $H = [B \mid I_{n-k}]$

3.3 Syndrome

Definition (Syndrome)**.** let H a PCM of a $[n, k]$-code C.
Then $\forall\ y \in V(n, q)$, the $1 \times (n-k)$ row vector

$$S(y) = yH^T$$

is called the *syndrome* of y.

Remark . i. for $h_1, h_2, \dots, h_{n-k}$ being rows of H, then

$$S(y) = (y \cdot h_1, y \cdot h_2, \dots, y \cdot h_{n-k})$$

ii. $S(y) = 0 \iff y \in C$

Intuition . Syndrome is the "sympthom" of the error. It ignores the valid part of the message and isolates the noise.

$$S(y) = yH^T = (c+e)H^T = \underbrace{cH^T}_{0} + eH^T = 0 + eH^T = eH^T$$

Syndrome allows the receiver to identify what went wrong without knowing what was originally sent.

Lemma 7.8. two vectors u, v are the same ocset of C iff they have the same syndrome.

Proof. if u, v in the same coset of C

$$\begin{aligned} &\implies u + C = v + C \\ &\implies u - v \in C \\ &\implies (u-v)H^T = 0 \\ &\implies uH^T = vH^T \\ &\implies S(u) = S(v) \end{aligned}$$

□

Corollary 7.9. There is a one-to-one ccorrespondence between cosets and syndromes.

$\longrightarrow$ for small n, it's fast to locate the received vector y in the array, but for large n: use syndrome to find which coset (ie. which row of the array) contains y.

Example 7.9. (from example 6.5)

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

by 7.6,

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

Hence the syndromes of the coset leaders are

$$S(0000) = 00$$
$$S(1000) = 11$$
$$S(0100) = 01$$
$$S(0010) = 10$$

The standard array becomes:

cosetleaders				*syndromes*
0000	1011	0101	1110	00
1000	0011	1101	0110	11
0100	1111	0001	1010	01
0010	1001	0111	1100	10

so, decoding algorithm: when a vector y is received, calculate $S(y) = yH^T$, locate $S(y)$ in the *syndromes* column of the array.

Locate y in the corresponding row and decode as the codeword at the top of the column containing y.

$\longrightarrow$ in practice: *syndrome lookup table*:

syndrome z	coset leader $f(z)$
00	0000
11	1000
01	0100
10	0010

decoding procedure:

1. receive y, calculate $S(y) = yH^T$
2. let $z = S(y)$, locate z in the first column of the looku table
3. decode y as $y - f(z)$

3.4 Incomplete decoding

For $d(C) = 2t + 1$ (or $2t + 2$), guarantee the correction of $\leq t$ errors in any codewordd and also detect some cases of more than t errors.

$\longrightarrow$ Arrange the cosets in order of increasing weight of the coset leaders.

Divide the array into a top part and a bottom part; top: cosets whose leaders have weights $\leq t$.

- If y in top part: decode as usual (assuming $\leq t$ errors).
- If y in bottom part: conclude that $> t$ errors have occurred, ask for retransmission.

3.5 Exercises

Exercise 7.1. prove Lemma 7.1

Proof. For any $u, v, w \in V(n, q)$ and $\lambda, \mu \in GF(q)$,

i. $u \cdot v = v \cdot u$

$$u \cdot v = \sum_{i=1}^{n} u_i \cdot v_i = \sum_{i=1}^{n} v_i \cdot u_i = v \cdot u$$

ii. $(\lambda u + \mu v) \cdot w = \lambda(u \cdot w) + \mu(v \cdot w)$

$$\begin{aligned}
(\lambda u + \mu v)w &= \sum(\lambda u_i + \mu v_i)w_i \\
&= \sum(\lambda u_i w_i + \mu v_i w_i) \\
&= \sum(\lambda u_i w_i) + \sum(\mu v_i w_i) \\
&= \lambda \sum(u_i w_i) + \mu \sum(v_i w_i) \\
&= \lambda \cdot (u \cdot w) + \mu \cdot (v \cdot w)
\end{aligned}$$

□

Exercise 7.2. prove that if E_n is the binary even weight code of length n, then $E_n^{\perp}$ is the repetition code of length n.

Proof. from exercise 5.2:

$$E_n = \{x \in V(n, 2) \mid w(x) \text{ is even}\} = \{(x_1, \ldots, x_n) \in F_2^n \mid x_1 + \ldots + x_n = 0\}$$

GM of E_n:

$$G = \left[\begin{array}{c|c} & 1 \\ I_{n-1} & \vdots \\ & 1 \end{array}\right]$$

thus, by 7.6, PCM is $H = \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}$.

If H is a PCM for C, then the rows of H generate $C^{\perp}$.
Therefore

$$E_n^{\perp} =< \underbrace{[1\ 1\ \cdots\ 1]}_{generator} >$$

then it's codewords are:

$$\begin{aligned} 0 \cdot (1, \ldots, 1) &= (0, \ldots, 0) \\ 1 \cdot (1, \ldots, 1) &= (1, \ldots, 1) \end{aligned}$$

$$\implies E_n^{\perp} = \{00\ldots00,\ 11\ldots1\}$$

ie. the repetition code of length n. ☐

Exercise 7.4. for a binary linear code with PCM H, show that the transpose of the syndrome of a received vector is equal to the sums of those columns of H corresponding to where the errors occurred.

Proof. Let e the error, y the vector received, and x the original codeword. Then $y = x + e$.

Then

$$S(y) = yH^T = (x+e)H^T = \underbrace{xH^T}_{\substack{0,\text{since } H \\ \text{is PCM of } C \\ \text{and } x \in C}} + eH^T = eH^T$$

So,

$$S(y)^T = e^T H = \sum_{j=1}^{n} e_j \cdot H_j$$

where H_j is the j-th column of H. ☐

Exercise 7.6. Let C be the ternary linear code with GM

$$G = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 1 \end{bmatrix}$$

(a) Find a generator matrix in standard form.

(b) Find a parity check matrix (PCM) for C in standard form.

(c) Use syndrome decoding to decode the received vectors: $2122, 1201, 2222$

Proof. (a)

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

thus $G = \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$ is PCM in standard form.

(b) since $G = \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$ then

$$H = \begin{bmatrix} -2 & -2 & 1 & 0 \\ -2 & -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

(c)

$$S(y) = yH^T = S(y) = (y \cdot h_1,\ y \cdot h_2).$$

w/ h_i the rows of H.

compute the lookup table (syndrome table):

e	$S(e)$
0000	00
1000	11
2000	22
0100	12
0200	21
0010	10
0020	20
0001	01
0002	02

left: weight-1 errors, since we're over F_3; right: coset leaders for the 9 cosets.

i. $y = 2121$:

$$S(y) = \begin{bmatrix} 2 & 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = [5, 5] \equiv [2, 2].$$

$S(y) = [2, 2]$, corresponds (in the table) to $e = 2000$.

$$\implies c = y - e = 2121 - 2000 = 0121$$

ii. $y = 1201$

$$S(y) = \begin{bmatrix} 1 & 2 & 0 & 1 \end{bmatrix} H^T = [3, 6] = [0, 0].$$

corresponds to $e = 0000$

$$\implies c = y - e = 1201$$

iii. $y = 2222$

$$S(y) = \begin{bmatrix} 2 & 2 & 2 & 2 \end{bmatrix} H^T = [6, 8] = [0, 2].$$

corresponds to $e = 0002$

$$\implies c = y - e = 2222 - 0002 = 2220$$

$\square$

4 Hamming Codes

Definition (Binary Hamming Code)**.** let r a positive integer, H a $r \times (2^r - 1)$ matrix whose columns are the distinct non-zero vectors of $V(r, 2)$.

The code having H as its PCM is called a *binary Hamming code*, denoted $Ham(r, 2)$.

Remark . $Ham(r, 2)$ has length $n = 2^r - 1$ and dimension $k = n - r$.

Redundancy of the code: $r = n - k$, the number of check symbols in each codeword.

Exercise 8.1. i. $r = 2$: $H = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

by Theorem 7.6, $G = [1\ 1\ \underbrace{1}_{I}]$

so $Ham(2, 2)$ is just the binary triple repetition code.

ii. $r = 3$: a PCM for $Ham(3, 2)$ is

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

(have taken columns in the natural order of increasing binary numbers from 1 to 7)

H in standard form: rearrange columns:

$$H = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

by Theorem 7.6, a generator matrix for $Ham(3, 2)$ is

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Theorem 8.2. the binary Hamming code $Ham(r, 2)$, for $r \geq 2$,

1. is a $[2^r - 1, 2^r - 1 - r]$-code

2. has minimum distance 3 (hence is single-error-correcting)

3. is a *perfect code*

Lemma (Decoding with a binary Hamming code)**.** $Ham(r, 2)$ is a perfect single-error-correcting code $\Longrightarrow$ the coset leader are precisely the $2^r \ \ (= n+1)$ vectors of $V(n, 2)$ of weight ≤ 1.

Let j-th column of H be the binary representation of j. Decoding algorithm:

1. receive vector y, calculate its syndrome: $S(y) = yH^T$

2. if $S(y) = 0$, assume $y = c$ (original sent codeword)

3. if $S(y) \neq 0$, assuming a single error, $S(y)$ gives the binary representation of the error position, and the error can be corrected.

Example . $H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$, if $y = 1101011$ then

$$S(y) = y \cdot H^T = 110$$

$\Longrightarrow$ 6-th position, and so we decode y as 1101001

4.1 Extended binary Hamming codes

$\hat{Ham}(r, 2)$, obtained from $Ham(r, 2)$ by adding an overall parity-check.

Not ideal for complete decoding, but good for incomplete decoding (correct any single error and detect any double error).

Let H be a PCM for $Ham(r, 2)$. A PCM $\hat{H}$ for $\hat{Ham}(r, 2)$:

$$\hat{H} = \begin{bmatrix} & & & 0 \\ & H & & \vdots \\ & & & 0 \\ 1 & \cdots & 1 & 1 \end{bmatrix}$$

If H is taken with columns in increasing order of binary numbers, we have a neat incomplete decoding algorithm:

Example 8.3. $r = 3$, $\hat{Ham}(3, 2)$. Received y, $S(y) = y\hat{H}^T = (s_1, s_2, s_3, s_4)$

i. if $s_4 = 0$ and $(s_1, s_2, s_3) = 0$: assume no errors

ii. if $s_4 = 0$ and $(s_1, s_2, s_3) \neq 0$: assume ≥ 2 errors and ask retransmission

iii. if $s_4 = 1$ and $(s_1, s_2, s_3) = 0$: assume 1 error in the last position

iv. if $s_4 = 1$ and $(s_1, s_2, s_3) \neq 0$: assume 1 error in the j-th position, where j is the number binary-represented by (s_1, s_2, s_3)

Theorem 8.4. suppose C a linear $[n, k]$-code over $GF(q)$ with PCM H. Minimum distance of C is d iff any $d-1$ columns of H are linearly independent but some d columns are linearly dependent.

Proof. by 5.2, min. distance of C (ie. $d(C)$) = smallest of the weights of the non-zero codewords.

Let $x = x_1 x_2 \dots x_n \in V(n, q)$, then

$$\begin{aligned} x \in C &\iff xH^T = 0 \\ &\iff x_1 H_1 + x_2 H_2 + \dots + x_n H_n = 0 \end{aligned}$$

where H_i denote the columns of H.

$\implies$ each codeeword x of weight d corresponds with a set of d linearly dependent columns of H.

On the other hand, if $\exists$ a set of $d-1$ lin.dependent columns of H, say $H_{i_1}, H_{i_2}, \dots, H_{i_{d-1}}$, then $\exists$ $x_{i_1}, \dots, x_{i_{d-1}}$ scalars, not all zero, such that

$$x_{i_1} H_{i_1} + x_{i_2} H_{i_2} + \dots + x_{i_{d-1}} H_{i_{d-1}} = 0$$

But then, the vector

$$x = (0 \dots 0 x_{i_1} 0 \dots x_{i_2} 0 \dots 0 x_{i_{d-1}} 0 \dots 0)$$

having x_{i_j} in the j-th position for $j \in [d-1]$, would satisfy $xH^T = 0$, and so would be a non-zero codeword of weight $< d$. $\square$

4.2 q-ary Hamming codes

For C a inear code with $d(C) = 3$ (min.dist), any 2 columns of a PCM H must be linearly independent

$\implies$ columns of H must be non-zero, and no column can be a scalar multiple of another

Any $0 \neq v \in V(r, q)$ vector has $q-1$ non-zero scalar multiples, forming the set

$$\{\lambda v \mid \lambda \in GF(q), \lambda \neq 0\}$$

$\longrightarrow$ the $q^r - 1$ non-zero vectors of $V(r, q)$ may be partitioned into $\frac{q^r-1}{q-1}$ such sets, called *classes.*

$\longrightarrow$ two vectors are scalar multiples of each other if they are in the same class

By choosing one vector from each class, obtain a set of $\frac{q^r-1}{q-1}$ vectors, where not two are linearly dependent.

By Theorem 8.4, taking these as columns of h gives a PCM for a

$$\left[\frac{q^r-1}{q-1}, \frac{q^r-1}{(q-1)-r}, 3\right]\text{-code}$$

$\Longrightarrow$ called *q-ary Hamming code*, $Ham(r,q)$.

$\longrightarrow$ PCM is unique up to permutation of columns

$\Longrightarrow$ Hamming codes are linear codes

Theorem 8.6. $Ham(r,q)$ is a perfect single-error-correcting code.

Proof. $Ham(r,q)$ is constructed to be a $(n,M,3)$-code, with $n = \frac{q^r-1}{q-1}$ and $M = q^{n-r}$.

With $t = 1$, the LHS of the sphere-packing bound becomes

$$q^{n-r} \cdot (1 + n(q-1)) = q^{n-r} \cdot (1 + q^r - 1) = q^n$$

which is the RHS, so $Ham(r,q)$ is a perfect code. $\square$

Decoding q-ary Hamming code Since Ham is perfect single-error correcting code, the coset leaders, other than 0, are the vectors of weight 1.

Syndrome of such coset leader:

$$S(0\dots0\ b\ 0\dots0) = (0\dots0\ b\ 0\dots0)\cdot H^T = b \cdot H_j^T$$

where H_j is the j-th column of H, and b is at the j-th position of y.

Decode:

- given received y, calculate $S(y) = yH^T$
- if $S(y) = 0$, assume no errors
- if $S(y) \neq 0$, then $S(y) = b \cdot H_j^T$ for some b and j
  correct by: subtract b from the j-th entry of y.

Example:

$$q = 5,\ H = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 3 & 4 \end{bmatrix},\ y = 203031$$

Then,

$$\begin{aligned} S(y) = y \cdot H^T &= 2 \cdot H_1^T + 3 \cdot H_3^T + 3 \cdot H_5^T + 1 \cdot H_6^T \\ &= 2 \cdot [0\ 1] + 3 \cdot [1\ 1] + 3 \cdot [1\ 3] + 1 \cdot [1\ 4] \\ &= [0\ 2] + [3\ 3] + [3\ 9] + [1\ 4] \\ &= [7\ 8] = [2\ 3] = \underbrace{2}_{b} \cdot \underbrace{[1\ 4]}_{H_6^T} \pmod 5 \end{aligned}$$

$\Longrightarrow$ error is in position 6-th, with magnitude $b = 2$

$\Longrightarrow$ subtract b from the j-th entry of y: $1 - 2 \pmod 5 = -1 \pmod 5 = 4$, therefore $y' = 203034$.

5 MLCT - Main Linear Coding Theory problem

A good (n, M, d)-code has

- small n, for fast transmission of messages
- large M, to enable transmission of a wide variety of messages
- large d, to correct many errors

Denote by $A_q(n, d)$ the largest M such that $\exists$ a q-ary (n, M, d)-code.

Theorem 2.7. Suppose d is odd. Then a binary (n, M, d)-code exists iff a binary $(n+1, M, d+1)$-code exists.

Proof. $\square$

Corollary 2.8. If d is odd, then $A_2(n+1, d+1) = A_2(n, d)$.
Equivalently, if d is even, then $A_2(n, d) = A_2(n-1, d-1)$.

Definition (Sphere)**.** $\forall\ u \in F_q^n$ (vector) and any integer $r \geq 0$, the *sphere* of radius r and centre u, denoted $S(u, r)$, is the set

$$\{v \in F_q^n \mid d(u, v) \leq r\}$$

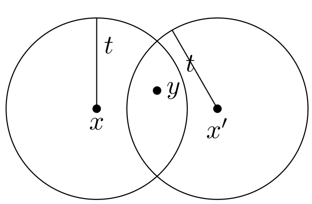
Theorem 1.9 & 2.12. a code C can

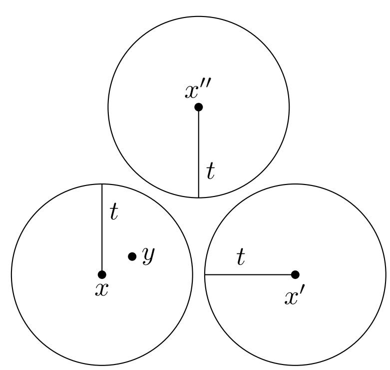
i. detect up to s errors in any codeword if $d(C) \geq s + 1$

ii. correct up to t errors in any codeword if $d(C) \geq 2t + 1$

  This is, the spheres of radius t centered on the codewords of C are disjoint (ie. have no overlap).

  So, if $\leq t$ errors occur, then the received vector y may be different from the sent codeword x, hence being different from the centre of the sphere $S(x, t)$, but can not 'escape' from the sphere, so is 'drawn back' to x by nearest neighbour decoding.





Definition (Perfect codes)**.** A code which achieves the sphere-packing bound is called a *perfect code*. Thus, for a perfect t-error-correcting code, the M spheres of radius t centred on codewords 'fill' the whole space F_q^n without overlapping.

In other words, every vector in F_q^n is at distance $\leq t$ from exactly one codeword.

Theorem (MLCT problem (Version 1))**.** For given length n, find the maximum dimension k such that $\exists$ an $[n, k, d]$-code over $GF(q)$. (Then, for this $k, B_q(n, d) = q^k$).

Remark . *redundancy* r of an $[n, k, d]$-code is just $r = n - k$ (the number of check symbols in a codeword).

Theorem (MLCT problem (Version 2))**.** For given redundancy r, find the maximum length n such that $\exists$ an $[n, n - r, d]$-code over $GF(q)$.

Definition ((n, s)-set)**.** a (n, s)-set in $V(r, q)$ is a set of n vectors in $V(r, q)$ with the property that any s of them are linearly independent.

Denote $max_s(r, q)$ the largest value of n for which $\exists$ a (n, s)-set in $V(r, q)$.

Definition (Packing problem)**.** The *packing problem* for $V(r, q)$ is that of determining the values of $max_s(r, q)$ and the optimal (n, s)-sets.

Theorem 14.1. There exists an $[n, n-r, d]$-code over $GF(q)$ iff $\exists$ an $(n, d-1)$-set in $V(r, q)$.

Proof. let C a $[n, n - r, d]$-code over $GF(q)$ with PCM H. Then by Theorem 8.4, the columns of H form a $(n, d - 1)$-set in $V(r, q)$.

On the other hand, let K a $(n, d-1)$-set in $V(r, q)$. If we form a $r \times n$ matrix H with the vectors of K as its columns, by Theorem 8.4, H is the PCM of a $[n, n - r]$-code whose minimum distance is at least d. □

Corollary 14.2. for given values q, d, r, the largest value of n for which there exists a $[n, n - r, d]$-code over $GF(q)$ is $max_{d-1}(r, q)$.

$\Longrightarrow$ the MLCT problem (V2) is the same as the packing problem of finding $max_{d-1}(r, q)$.

5.1 The projective geometry PG(r-1,q)

Let the vector space

$$V(r, q) = \{(a_1, a_2, \ldots, a_r) \mid a_i \in GF(q)\}$$

Definition (Projective Geometry)**.** $PG(r - 1, q)$

- *points* are the one-dimensional subspaces of $V(r, q)$

- *lines* are the two-dimensional subspaces of $V(r,q)$

Each point $P \in PG(r-1,q)$, as a subspace of $V(r,q)$ of dimension 1, is generated by a single non-zero vector.

So if $a = (a_1, a_2, \ldots, a_r) \in P$, then $P = \{\lambda a \mid \lambda \in GF(q)\}$.

$\Longrightarrow$ for $a = (a_1, a_2, \ldots, a_r),\ b = (b_1, b_2, \ldots, b_r) \in P$, then

$$a = b \text{ in } PG(r-1,q) \text{ iff } a = \lambda b \text{ in } V(r,q)$$

for some non-zero scalar λ.

Lemma 14.6. in $PG(r-1,q)$,

1. number of points is $\frac{q^r-1}{q-1}$
2. any two points lie on exactly one line
3. each line contains exactly $q+1$ points
4. each point lies on $\frac{q^{r-1}-1}{q-1}$ lines

Proof.

1. number of points is $\frac{q^r-1}{q-1}$:
since each of the q^r-1 nonzero vectors in $V(r,q)$ has $q-1$ nonzero scalar multiples, the number of points of $PG(r-1,q)$ is $\frac{q^r-1}{q-1}$
2. any two points lie on exactly one line:
if a, b distinct points of $PG(r-1,q)$, then the unique line through them consists of the points $\lambda a + \mu b$, where λ, μ are scalars not both zero.
3. each line contains exactly $q+1$ points:
in (ii) there are q^2-1 choices for the pair (λ, μ), but we're identifying scalar multiples, so the number of distinct points on the line is
$$\frac{q^2-1}{q-1} = q+1$$
4. each point lies on $\frac{q^{r-1}-1}{q-1}$ lines:
let t be the number of lines on which a point P lies. Let the set
$$X = \{(Q, L) \mid Q \text{ is a point } \neq P,\ L \text{ is a line containing both } P \text{ and } Q\}$$
Count the members of X in two ways:
for each of the $\frac{q^r-1}{q-1} - 1$ choices of Q, $\exists!$ line containing P and Q. Thus,
$$|X| = \frac{q^r-1}{q-1} - 1 = \frac{q^r-q}{q-1}$$
On the other hand, for each of the t lines through P, there are (by (iii)) q points Q other than P lying on L.

Thus, $|X| = tq$,

$$\Longrightarrow tq = \frac{q^r - q}{q-1}, \qquad \Longrightarrow t = \frac{q^{r-1} - 1}{q-1}$$

$\square$

Definition (Projective plane)**.** The $PG(2, q)$ is called the *projective plane* over $GF(q)$.

References

[1] Raymond Hill. A First Course in Coding Theory, 1986.

[2] J.H. van Lint. Introduction to Coding Theory, 1981.